25900245

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBTS200 - CLINICAL BIOCHEMISTRY AND CLINICAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours ` Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall

outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** mark

1. Body Mass Index (BMI) above **30 or higher** indicates obesity
2. Common cold is primarily transmitted through **respiratory droplets**
3. **Blood** smear is the gold standard method for malaria diagnosis.
4. The **cerebellum** is responsible for balance, posture, and coordination of movements.
5. In the Jaffe's reaction, creatinine reacts with **picric acid** in alkaline medium to form a red-orange color.
6. The **left atrium** receives oxygenated blood from the lungs.
7. The kidney converts inactive vitamin D into its active form **calcitriol**

8. Amoebiasis is more common in regions with poor **hygiene** and sanitation

9. Salmonella Typhimurium and Salmonella Enteritidis commonly cause **gastroenteritis** infection

10. A common biochemical marker of lung injury is increased level of Angiopoietin-1 and -2 (Ang-1 and -2) in serum

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11. State any three major lifestyle factors that contribute to kidney damage.
Unhealthy habits like smoking, excessive alcohol consumption, and a diet high in salt and processed foods, as well as a sedentary lifestyle that leads to obesity, high blood pressure, and uncontrolled diabetes
Write any 3 lifestyle factors (**3** marks)

12. In what way can aflatoxin damage liver cells?
Metabolized into reactive epoxide compounds that bind to DNA and proteins, causing mutations, cell death, and oxidative stress
Mode of action (**3** marks)

13. What are the blood tests useful in diagnosing acute pancreatitis?
Elevated levels of digestive enzymes, lipase and amylase**.**

14. List two diagnostic tests used for coronavirus infection.
Nasal swab detection method, RT-PCR

15. How does alcohol affect cardiovascular health?
Cardiomyopathy

16. Name two laboratory diagnostic methods for filariasis.
Blood smear microscopy and antigen or antibody detection tests

17. State the role of the liver in detoxification with an example.
Filtration and bio transformation

18. Mention the normal range of total cholesterol in human blood.
Less than 200mg/DL

19. Name two laboratory diagnostic methods for oral thrush.
KOH (potassium hydroxide) microscopy and fungal culture

20. List three common symptoms of pulmonary tuberculosis.

Persistent cough lasting three weeks or longer, chest pain, and coughing up blood or bloody sputum

21. Explain the role of 24-hour urinary free cortisol in adrenal assessment
Assesses adrenal gland function by measuring the total amount of free cortisol excreted in urine over a 24-hour period

22. State three preventive measures of cholera.
Ensuring safe water, practicing good hygiene, and consuming safe food

{3 x 10 = 30}

Part C

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

23. Explain complications of RSV infection and their clinical significance.
    - RSV infection -1 mark
    - Complications -2 marks
    - Clinical significance -2 marks

24. Discuss the complications and prognosis of typhoid fever
    - Typhoid fever - 1 mark
    - Complications - 2 marks
    - Prognosis - 2 marks

25. What are the normal ranges of fasting and postprandial blood sugar, and how are they altered in diabetes mellitus?
    - Normal ranges of fasting and postprandial blood sugar- 2 marks
    - Changes observed in diabetes mellitus - 1 mark

26. Explain the symptoms of diabetes mellitus in detail.
    - Write any 5 symptoms in detail

27. What are five functions of calcitonin secreted by the thyroid gland
    - Write five functions -1 mark each

28. Explain the complications of invasive aspergillosis.
    - Invasive aspergillosis - 3 marks
    - Any 4 complications - 2 marks

29. Discuss the functions of the anterior pituitary hormones with examples.
    - Write any five hormones and their functions -1 mark each

30. Discuss the role of stress and lifestyle in cancer progression.
    - Introduction -1 mark
    - Steps in cancer progression -2 marks
    - Role of lifestyle and stress -2 marks

[5 x6 = 30]